# MTH 201 Infinite Series (Up through comparison tests)

1. Show how comparisons can be used to determine if the following series converge or diverge.

a) $\sum_{n=1}^{\infty} \frac{n-1}{n^2\sqrt{n}}$

b) $\sum_{n=1}^{\infty} \frac{1+\sin(n)}{10^n}$

c) $\sum_{n=1}^{\infty} \frac{n+4^n}{n+6^n}$

d) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

e) $\sum_{n=6}^{\infty} \frac{\sqrt{n^3+2n+1}}{3n^2+14n+2}$

f) $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$